Midterm 3

● Graded

Student

James Ursulica

Total Points

100 / 100 pts

Question 1

Question 1 — **30** / 30 pts

1.1 a — **10** / 10 pts

- ✔ **− 0 pts** Correct
- **− 2 pts** first is wrong
- **− 2 pts** second is wrong
- **− 2 pts** third is wrong
- **− 2 pts** fourth is wrong
- **− 2 pts** fifth is wrong

1.2 b — **10** / 10 pts

- ✔ **− 0 pts** Correct
- **− 2 pts** h1 should be 3093.9 kj/kg
- **− 0 pts** h1=h2
- **− 10 pts** not done or wrong
- **− 2 pts** interpolation is wrong
- **− 2 pts** T1 is not equal to T2

1.3 c — **10** / 10 pts

- ✔ **− 0 pts** Correct
- **− 6 pts** equation is wrong
- **− 1 pt** calculation error
- **− 10 pts** not done
- **− 4 pts** deltaT are wrong

Question 2

Question 2 **30** / 30 pts

2.1 a **4** / 4 pts

✔ **– 0 pts** Correct

– 2 pts not shell and tube

– 3 pts inlets and outlets are missing or wrong

– 4 pts not done

2.2 b **5** / 5 pts

✔ **– 0 pts** Correct

– 1 pt control volume

– 5 pts not done

2.3 c **6** / 6 pts

✔ **– 0 pts** Correct

– 3 pts mass balance missing or wrong (should be m1(dot)+m2(dot) = m3(dot)

– 2 pts simplified energy balance is wrong(should be 0=m1(dot)*h1+m2(dot)*h2-m3(dot)*h3

– 1 pt error in energy balance

– 6 pts not done

2.4 d **10** / 10 pts

✔ **– 0 pts** Correct

– 1 pt h1= 3188.4

– 1 pt h2= 251.13

– 1 pt h3= 2748.7

– 5 pts equation is wrong

– 1 pt m1(dot)=333.3 kg/s

– 10 pts not done or wrong

– 1 pt calculation error

2.5 e **5** / 5 pts

✔ **− 0 pts** Correct

− 1 pt calculation error

− 3 pts Equation is wrong (m1(dot)+m2(dot)=m3(dot))

− 5 pts not done

Question 3

Question 3 **40** / 40 pts

3.1 a **5** / 5 pts

✔ **− 0 pts** Correct

− 5 pts not done

3.2 b **3** / 3 pts

✔ **− 0 pts** Correct

− 1 pt closed system

− 1 pt ideal gas

− 1 pt not control volume

− 3 pts not done

3.3 c **5** / 5 pts

✔ **− 0 pts** Correct

− 3 pts closed system energy balance

− 5 pts not done

3.4 d **10** / 10 pts

✔ **− 0 pts** Correct

− 2 pts COP is wrong (COP= Th/(Th-Tc) = 2)

− 2 pts Qh= 250*1.5 =375

− 2 pts COP= Qh/W 2=375/W thus W=187.5

− 10 pts not done

3.5 e **10** / 10 pts

✔ **− 0 pts** Correct

− 10 pts not done or wrong

− 2 pts T=300K

− 1 pt P2=76.1kPa

− 7 pts equation is right, but not done

3.6 f **7** / 7 pts

✔ **− 0 pts** Correct

− 7 pts not done or wrong

− 2 pts adibatic parts, specific volume is not the same

− 2 pts isothermal parts, P is not the same

CHE 201 MidTerm Exam 3

November 24th, 2025

NAME: James Ursulica

Please fill in answers in the space supplied and show all your work to receive full credit.

1. a) (10 pts) State whether the following statements are **True of False**

T A *diffuser* is a flow passage of varying cross-sectional area in which the pressure of a gas or liquid increases in the direction of flow.

T Enthalpy for the inlet of the turbine is greater than the outlet of the turbine. $\dot{W} = \dot{m}(h_i - h_e)$ +

F A pump is an equipment to increase the pressure or to elevate gas.

F Turbine is an example of steady flow device that result in an increase in working fluid pressure from inlet to exit.

T The second Carnot corollary states that reversible power cycles operating between the same two thermal reservoirs have the same thermal efficiency.

b) (10 pts) Steam at 1MPa and 320°C is throttled adiabatically to a pressure of 0.7 bar. If the change in kinetic energy is negligible, calculate the temperature and specific volume of the steam after throttling.

$\bowtie$ $\quad P_i = 1\,MPa = 10\,bar \quad P_e = 0.7\,bar$

$h_i = h_e \quad T_i = 320°C$

$T_e(0.7\,bar, h_e = 3093.9\,\frac{kJ}{kg}) \Rightarrow$ interpolate

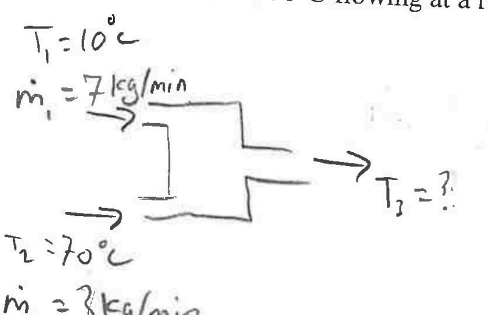
$h_i(\text{steam}, 10\,bar, 320°C) = 3093.9\,\frac{kJ}{kg} = h_e$

$$\frac{320 - 280}{3115.3 - 3035} = \frac{T_e - 280}{3093.9 - 3035}$$

$$v_e: \frac{3.905 - 3.640}{320 - 280} = \frac{v_e - 3.640}{309.3 - 280}$$

$$\boxed{v_e = 3.83\,\frac{m^3}{kg}} \quad \boxed{T_e = 309.3\,°C}$$

c) (10 points) In a shower, cold water at 10°C flowing at a rate of 7kg/min is mixed with hot water at 70°C flowing at a rate of 3kg/min. Calculate the exit temperature of the mixture.



$\dot{Q} = -\dot{Q}$

$\dot{m}_1 c \Delta T = \dot{m}_2 c \Delta T$

$(\frac{7kg}{min})(T_f - 10°C) = -(\frac{3kg}{min})(T_f - 70°C)$

$7T_f - 70 = -3T_f + 210$

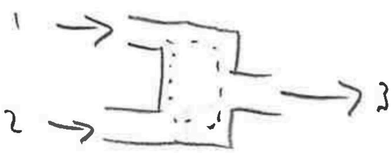
$10T_f = 280$

$$\boxed{T_f = 28°C}$$

2. (30 pts)
A stream of superheated steam at 5 bar and 360 °C with a mass flow rate of 20,000 kg/min (stream 1) is mixed with a stream of liquid water at 5 bar and 60 °C (stream 2) in an adiabatic mixing chamber. The resulting mixture exits as a saturated vapor at 5 bar (stream 3). Assume steady state operation and neglect any changes in kinetic and potential energy.

a) (4 pts) Draw a schematic of the system and label the system boundary.
b) (5 pts) State the engineering model.
c) (6 pts) Write the simplified balances for mass and energy .
d) (10 pts) Calculate the mass flow rate of the liquid water entering the mixing chamber, in kg/s.
e) (5 pts) Calculate the mass flow rate of the saturated vapor exiting the mixing chamber, in kg/s.

a)



b) - control volume
- $\Delta KE = \Delta PE = 0$
- adiabatic
- Steady state
- 2 inlet 1 outlet

$\dot{m}_1 = 20000 \frac{kJ}{min} = 333.3 \frac{kg}{s}$

c) $\cancel{\frac{dm_{cv}}{dt}} = \sum \dot{m}_i - \sum \dot{m}_e$

$0 = \dot{m}_1 + \dot{m}_2 - \dot{m}_3$

$\boxed{\dot{m}_3 = \dot{m}_1 + \dot{m}_2}$

$\cancel{\frac{dE_{cv}}{dt}} = \cancel{\dot{Q}} - \cancel{\dot{W}} + \sum \dot{m}_i (h_i + \cancel{\frac{V_i^2}{2}} + \cancel{gz_i}) - \sum \dot{m}_e (h_e + \cancel{\frac{V_e^2}{2}} + \cancel{gz_e})$

$\boxed{0 = \dot{m}_1 h_1 + \dot{m}_2 h_2 - \dot{m}_3 h_3}$

$0 = \dot{m}_1 h_1 + \dot{m}_2 h_2 - (\dot{m}_1 + \dot{m}_2) h_3$

d)

$h_1(\text{steam}, 5\text{ bar}, 360°C) = 3188.4 \frac{kJ}{kg}$

$h_2(\text{Comp. Liq.}, 5\text{ bar}, 60°C) = h_f(T) = 251.13 \frac{kJ}{kg}$

$h_3(\text{Sat. Vap}, 5\text{ bar}) = 2748.7 \frac{kJ}{kg}$

$0 = \dot{m}_1 h_1 + \dot{m}_2 h_2 - \dot{m}_1 h_3 - \dot{m}_2 h_3$

$\dot{m}_2 (h_3 - h_2) = \dot{m}_1 (h_1 - h_3)$

$\dot{m}_2 = \dot{m}_1 (h_1 - h_3) / (h_3 - h_2)$

$\dot{m}_2 = \frac{(333.3 \frac{kg}{s})(3188.4 - 2748.7 \frac{kJ}{kg})}{2748.7 - 251.13 \frac{kJ}{kg}}$

$\boxed{\dot{m}_2 = 58.68 \frac{kg}{s}}$

e) $\dot{m}_3 h_3 = \dot{m}_1 h_1 + \dot{m}_2 h_2$

$\dot{m}_3 = (\dot{m}_1 h_1 + \dot{m}_2 h_2) / h_3$

$\dot{m}_3 = \frac{((333.3 \frac{kg}{s})(3188.4 \frac{kJ}{kg}) + (58.68 \frac{kg}{s})(251.13 \frac{kJ}{kg}))}{2748.7 \frac{kJ}{kg}}$

$\boxed{\dot{m}_3 = 392.02 \frac{kg}{s}}$

adiabatic comp → Isothermal Exponsion → adiabatic expansion → Isothermal Compres

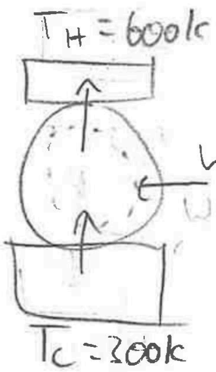
pump Boiler reversible Cooler

3. (40 points)

1.5 kg of Air within a piston-cylinder assembly executes a Carnot heat pump cycle. For the cycle, $T_H = 600$ K and $T_C = 300$ K. The energy rejected by heat transfer has a magnitude of 250 kJ per kg of air. The pressure at the start of the isothermal expansion is 325 kPa. Assuming the ideal gas model for the air, determine ~~(a)~~ the magnitude of the net work input, in kJ per kg of air, and ~~(b)~~ the pressure at the end of the isothermal expansion, in kPa.a) (5 pts)

a) (5 pts) Draw a schematic of the system and label the system boundary.
b) (3 pts) State the engineering model
c) (5 pts) Write the simplified energy rate balance
d) (10 pts) Determine the magnitude of the net work input, in kJ.
e) (10 pts) Determine the pressure at the end of the isothermal expansion, in kPa.
f) (7 pts) Sketch the process on a P-V diagram.

a)



b)
- Carnot cycle
  ⇒ reversible
- Ideal gas
- Isothermal Expansion
- closed system

d) $Cop = \frac{Q_H}{W}$ $Q_H = 250 \frac{kJ}{kg}$

$Cop = \frac{T_H}{T_H - T_L} = \frac{600}{600-300} = 2$

$2 = \frac{250}{W}$ $W = \frac{250}{2}$

$W = 125 \ kJ/kg (1.5 kg)$

$\boxed{W = 187.5 \ kJ}$

c)

$\Delta E = Q - W$

$\Delta U + \Delta KE + \Delta PE = Q - W$

$\boxed{\Delta U = Q - W}$

e) $Pv = RT$ $P_1 v_1 = P_2 v_2 = constant$

$W = \int P dv$

Isothermal Expansion, T = 300k

$W = \int \frac{constant}{v} dv$

$(325 kPa)(v_1) = (8.314 \frac{kJ}{kmol \, k})(\frac{1 \, kmol}{28.97 \, kg})(300k)$

$= P_1 v_1 \ln \frac{v_2}{v_1}$

$v_1 = 0.265 \frac{m^3}{kg}$

$125 = (325)(.265) \ln(\frac{v_2}{.265})$

$e^{\ln \frac{v_2}{.265}} = e^{1.45}$

$\frac{v_2}{\cancel{.265}} = e^{1.45}$ $v_2 = 1.132 \frac{m^3}{kg}$

e) $P_1 v_1 = P_2 v_2$

$(325\ \text{kPa})(0.265\ \frac{m^3}{kg}) = P_2 (1.132\ \frac{m^3}{kg})$

$$\boxed{P_2 = 76.1\ \text{kPa}}$$

f)